Assignment : Solving Differential eq^n using Finite Diff. Method

$$\frac{d^2u}{dx^2} = Kx \quad , \; 0 < x < 1$$

@ $x = 0$: $u = 0$

@ $x = 1$: $\frac{du}{dx} = 1$

$$\Delta x = \frac{1}{N}$$

$$x_i = i\Delta x \quad , \; i = 0, 1, \ldots, N$$

$$\frac{d^2u}{dx^2} = \frac{u_{i+1} - 2u_i + u_{i-1}}{(\Delta x)^2} + O(\Delta x)^2$$

Central Difference

$$\Rightarrow \frac{u_{i+1} - 2u_i + u_{i-1}}{(\Delta x)^2} = Kx_i = Ki\Delta x$$

$$\Rightarrow u_{i+1} - 2u_i + u_{i-1} = Ki(\Delta x)^3 \quad \text{—— (1)}$$

for $i = 1, \ldots, N-1$

from BC1, $u_0 = 0$

from BC2,

$$\frac{du}{dx} = \frac{u_{N+1} - u_{N-1}}{2\Delta x} + O(\Delta x)^2$$

$$\Rightarrow \frac{u_{N+1} - u_{N-1}}{2\Delta x} = 1$$

$$\Rightarrow u_{N+1} = u_{N-1} + 2\Delta x$$

Substituting this into (1),

$$2u_{N-1} - 2u_N + 2\Delta x = KN(\Delta x)^3$$

$\Rightarrow$

$$\begin{pmatrix} 1 & & & & & & \\ 1 & -2 & 1 & & & 0 & \\ & 1 & -2 & 1 & & & \\ & & 1 & -2 & 1 & & \\ & & & \ddots & \ddots & \ddots & \\ 0 & & & & 1 & -2 & 1 \\ & & & & & 2 & -2 \end{pmatrix} \begin{pmatrix} u_0 \\ u_1 \\ \vdots \\ \vdots \\ u_N \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ K(\Delta x)^3 \\ K2(\Delta x)^3 \\ \vdots \\ K(N-1)(\Delta x)^3 \\ KN(\Delta x)^3 - 2\Delta x \end{pmatrix}$$

$$\underset{(N+1)\times(N+1)}{A}\ \underset{(N+1)\times 1}{u} = \underset{(N+1)\times 1}{b} \Rightarrow \text{Solve}$$

for Analytical sol^n,

$$\frac{d^2u}{dx^2} = Kx \Rightarrow \frac{du}{dx} = \frac{Kx^2}{2} + c_1$$

$$\Rightarrow u = \frac{Kx^3}{6} + c_1 x + c_2$$

using BCs, $c_2 = 0$

$$\frac{K}{2} + c_1 = 1 \Rightarrow c_1 = 1 - \frac{K}{2}$$

$$\therefore u(x) = \frac{Kx^3}{6} + \left(1 - \frac{K}{2}\right)x$$